

COVERTRACE-SAT as Disjoint-Subcube Knowledge Compilation

con sus Extensiones: Fragmentación, Resolución Arbórea, Espectro de
Fourier y Representaciones Poliédricas

Oscar Riveros

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Part I

COVERTRACE-SAT as Disjoint-Subcube Knowledge Compilation: Worst-Case Fragmentation, Conditional PH Collapse, and Connections to Geometric Complexity Theory

COVERTRACE-SAT as Disjoint-Subcube Knowledge Compilation

Worst-Case Fragmentation, Conditional PH Collapse, and Connections to Geometric Complexity Theory

Oscar Riveros

<https://github.com/maxtuno/>

<https://independent.academia.edu/oarr>

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Compilación geométrica de SAT/#SAT: cada cláusula prohíbe un subcubo, y COVERTRACE mantiene la unión prohibida como familia disjunta para conteo exacto. Cota inferior exponencial para paridad; compilación DSOP uniforme $\Rightarrow$ PH = P. Extensión afín sobre $\mathbb{F}_2$ y perspectiva GCT.

Abstract

We study COVERTRACE-SAT, an exact SAT/#SAT algorithm that interprets a CNF formula as a union of forbidden axis-aligned subcubes in the Boolean hypercube $\Omega_n = \{0, 1\}^n$ and maintains this union as a disjoint family of subcubes. The disjointness invariant yields immediate exact model counting by volume additivity and enables constructive witness extraction in time polynomial in the representation size. We formalize the core algorithm, prove correctness, and analyze complexity in terms of fragmentation (the number of disjoint subcubes maintained). We establish a tight exponential worst-case lower bound: the odd-parity set in Ω_n requires 2^{n-1} disjoint axis-aligned subcubes, implying that COVERTRACE-style compilation can require exponential space/time on explicitly given inputs (of exponential description size; the polynomial-size case remains open, cf. Remark 6.4). We then place COVERTRACE within the knowledge compilation landscape: the maintained family is precisely a deterministic DNF / disjoint sum of products (DSOP) for $\neg F$. This yields a clean conditional consequence: if every CNF admitted a uniform polynomial-time compilation into a polynomial-size disjoint-subcube cover of its forbidden region, then #SAT $\in$ P and the Polynomial-Time Hierarchy collapses to P (via Toda's theorem). Finally, motivated by parity-like obstructions, we propose an affine extension (subspaces over $\mathbb{F}_2$) that compresses parity exponentially, and we articulate a GCT-inspired obstruction program by viewing disjoint subcubes as restricted rank-one tensors. We emphasize that lower bounds for a specific representation language do not constitute a proof of P $\neq$ NP, but they do suggest concrete, mathematically precise barriers and open problems.

1 Introduction

COVERTRACE-SAT is an exact algorithmic viewpoint on SAT and #SAT based on an elementary geometric duality: every clause of a CNF forbids exactly the assignments that falsify it, and those assignments form an axis-aligned subcube of the Boolean hypercube $\Omega_n = \{0, 1\}^n$.

If F is a CNF formula, let $U(F) \subseteq \Omega_n$ denote the set of assignments that falsify at least one clause. Then

$$\#\text{SAT}(F) = 2^n - |U(F)|.$$

COVERTRACE-SAT incrementally constructs a disjoint family of forbidden subcubes whose union is exactly $U(F)$. This disjointness invariant transforms inclusion–exclusion into exact additivity. The price of exactness is fragmentation: adding a new forbidden cube to a disjoint union may force splitting existing cubes into many pieces. This paper provides a rigorous treatment of this phenomenon and connects it to three broader frameworks:

1. Knowledge compilation: the family maintained by COVERTRACE is a deterministic DNF / disjoint sum of products (DSOP) for $\neg F$.
2. Complexity theory: a hypothetical uniform polynomial-time compilation of arbitrary CNFs to polynomial-size DSOP would imply $\#\text{SAT} \in \text{P}$ and collapse the Polynomial-Time Hierarchy (PH).
3. Geometric Complexity Theory (GCT): disjoint subcubes correspond to a restricted class of rank-one tensors; fragmentation lower bounds become restricted tensor-rank lower bounds, suggesting a discrete “obstruction” viewpoint.

What this paper is not. Despite the geometric language, we do not claim any proof of $\text{P} \neq \text{NP}$. We provide (i) exact correctness results, (ii) unconditional worst-case lower bounds for a specific representation language, and (iii) conditional collapses under strong uniform compilation hypotheses. Bridging such lower bounds to $\text{P} \neq \text{NP}$ would require universal lower bounds against all polynomial-time algorithms, far beyond a single representation.

2 Geometric semantics of CNF

2.1 Patterns and subcubes

Definition 2.1 (Patterns and subcubes). A pattern is a vector $p \in \{0, 1, \bullet\}^n$, where $\bullet$ denotes “free”. The subcube induced by p is

$$Q(p) = \{x \in \Omega_n : \forall i \in [n], p_i \neq \bullet \Rightarrow x_i = p_i\}.$$

The support is $\text{supp}(p) = \{i : p_i \in \{0, 1\}\}$ and the width is $k(p) = |\text{supp}(p)|$. The volume is $\text{vol}(p) = |Q(p)| = 2^{n-k(p)}$.

Definition 2.2 (Disjoint families). A family $U \subseteq \{0, 1, \bullet\}^n$ is disjoint if $Q(u) \cap Q(v) = \emptyset$ for all distinct $u, v \in U$. Its covered set is $\text{Cov}(U) = \bigcup_{u \in U} Q(u)$.

2.2 Clauses induce forbidden subcubes

Let F be a CNF formula on variables $x_1, \dots, x_n$ with clauses $C_1, \dots, C_m$.

Definition 2.3 (Clause-to-cube map). For a clause C , define the pattern $p(C) \in \{0, 1, \bullet\}^n$ coordinate-wise:

$$p(C)_i = \begin{cases} 0, & \text{if } x_i \in C, \\ 1, & \text{if } \neg x_i \in C, \\ \bullet, & \text{if } x_i \notin C \text{ and } \neg x_i \notin C. \end{cases}$$

Remark 2.4 (Tautological clauses). If a clause C contains both x_i and $\neg x_i$, then $p(C)$ would be forced to take two values on coordinate i , and no pattern represents the (empty) falsification set of C . Henceforth all clauses are assumed non-tautological: a tautological clause is satisfied by every assignment, forbids nothing, and may be dropped from F without changing $U(F)$.

Proposition 2.5 (Falsifying assignments form a subcube). *An assignment $x \in \Omega_n$ falsifies C if and only if $x \in Q(p(C))$.*

Definition 2.6 (Forbidden region). The forbidden region of F is

$$U(F) = \bigcup_{j=1}^m Q(p(C_j)).$$

Then $\#SAT(F) = 2^n - |U(F)|$ and F is satisfiable iff $U(F) \neq \Omega_n$.

3 The COVERTRACE core algorithm

3.1 Elementary cube relations

We use two fundamental relations on patterns $p, r \in \{0, 1, \bullet\}^n$.

- Intersection: $Q(p) \cap Q(r) \neq \emptyset$ iff p and r do not disagree on any jointly-fixed coordinate.
- Containment: $Q(p) \subseteq Q(r)$ iff every coordinate fixed by r is also fixed by p with the same value.

3.2 Disjoint difference CubeDiff

The engine of COVERTRACE is a disjoint decomposition of a cube difference.

Problem 3.1 (CubeDiff). Given patterns $p, r \in \{0, 1, \bullet\}^n$, compute a disjoint family $D \subseteq \{0, 1, \bullet\}^n$ such that

$$Q(p) \setminus Q(r) = \bigsqcup_{d \in D} Q(d),$$

where $\sqcup$ denotes disjoint union.

A standard constructive solution splits p along a coordinate fixed in r but free in p .

Algorithm 1 CubeDiff(p, r): disjoint decomposition of $Q(p) \setminus Q(r)$

Require: $p, r \in \{0, 1, \bullet\}^n$

Ensure: disjoint family D with $\bigsqcup_{d \in D} Q(d) = Q(p) \setminus Q(r)$

```

1: if  $Q(p) \cap Q(r) = \emptyset$  then
2:   return  $\{p\}$ 
3: end if
4: if  $Q(p) \subseteq Q(r)$  then
5:   return  $\emptyset$ 
6: end if
7: choose an index  $i$  such that  $p_i = \bullet$  and  $r_i \in \{0, 1\}$ 
8: let  $b \leftarrow r_i$ 
9: define  $p_{\neq}$  by fixing  $p_i \leftarrow 1 - b$ 
10: define  $p_{=}$  by fixing  $p_i \leftarrow b$ 
11: return  $\{p_{\neq}\} \cup \text{CubeDiff}(p_{=}, r)$ 

```

Lemma 3.2 (Size bound for a single difference). *Let $p, r \in \{0, 1, \bullet\}^n$ with $Q(p) \cap Q(r) \neq \emptyset$ and $Q(p) \not\subseteq Q(r)$. Then $\text{CubeDiff}(p, r)$ returns at most $|\text{supp}(r) \setminus \text{supp}(p)|$ subcubes, and runs in $O(n \cdot |\text{supp}(r) \setminus \text{supp}(p)|)$ time in a direct pattern representation.*

Proof. Each recursive step fixes one coordinate i that is fixed in r but free in p . Thus the recursion depth is at most $|\text{supp}(r) \setminus \text{supp}(p)|$. Exactly one cube is emitted per recursion level (the branch $p_{\neq}$), and the recursion terminates when the remaining branch becomes a subset of r . $\square$

3.3 Incremental disjointization of the forbidden union

The next procedure adds a new forbidden cube Q into an existing disjoint family U by disjointizing Q against all existing cubes.

Algorithm 2 AddCube(U, q): add $Q(q)$ to a disjoint family

Require: disjoint family $U \subseteq \{0, 1, \bullet\}^n$, pattern $q \in \{0, 1, \bullet\}^n$

Ensure: disjoint family U' with $\text{Cov}(U') = \text{Cov}(U) \cup Q(q)$ and the added volume Δ

```

1:  $P \leftarrow \{q\}$  {pieces of  $Q(q)$  not yet subtracted}
2: for each  $R \in U$  do
3:    $P_{\text{new}} \leftarrow \emptyset$ 
4:   for each  $p \in P$  do
5:      $P_{\text{new}} \leftarrow P_{\text{new}} \cup \text{CubeDiff}(p, R)$ 
6:   end for
7:    $P \leftarrow P_{\text{new}}$ 
8:   if  $P = \emptyset$  then
9:     return  $(U, 0)$ 
10:  end if
11: end for
12:  $\Delta \leftarrow \sum_{p \in P} 2^{n-k(p)}$ 
13:  $U' \leftarrow U \cup P$  {disjoint union since  $P$  is outside  $\text{Cov}(U)$ }
14: return  $(U', \Delta)$ 

```

3.4 COVERTRACE-SAT

Algorithm 3 COVERTRACE-SAT (core, no heuristics)

Require: CNF $F = \bigwedge_{j=1}^m C_j$ over n variables

Ensure: UNSAT if F is unsatisfiable; otherwise SAT with a witness assignment

```

1:  $U \leftarrow \emptyset$ ;  $y \leftarrow 2^n$  { $y$  tracks surviving assignments}
2: for  $j = 1, \dots, m$  do
3:    $q \leftarrow p(C_j)$ 
4:    $(U, \Delta) \leftarrow \text{AddCube}(U, q)$ 
5:    $y \leftarrow y - \Delta$ 
6:   if  $y = 0$  then
7:     return UNSAT
8:   end if
9: end for
10: extract some  $x \in \Omega_n \setminus \text{Cov}(U)$ 
11: return SAT with witness  $x$ 

```

4 Correctness and exact model counting

Theorem 4.1 (Invariant and correctness). *After processing clauses $C_1, \dots, C_t$, the family U_t is disjoint and satisfies*

$$\text{Cov}(U_t) = \bigcup_{j=1}^t Q(p(C_j)).$$

Moreover,

$$y_t = 2^n - |\text{Cov}(U_t)| = \#\text{SAT}\left(\bigwedge_{j=1}^t C_j\right).$$

Hence COVERTRACE-SAT outputs UNSAT iff F is unsatisfiable, and otherwise outputs SAT with at least one satisfying assignment.

Proof. We proceed by induction on t . For $t = 0$, $U_0 = \emptyset$ and the claim holds. Assume the claim for $t - 1$. At step t , $\text{AddCube}(U_{t-1}, q)$ with the pattern $q = p(C_t)$ computes P , a disjoint family whose union is exactly $Q(q) \setminus \text{Cov}(U_{t-1})$. Subtracting sequentially over all $R \in U_{t-1}$ computes a disjoint decomposition of $Q(q) \setminus \text{Cov}(U_{t-1})$ because each call to CubeDiff maintains a disjoint decomposition of a difference. Since P is disjoint from $\text{Cov}(U_{t-1})$, the union $U_t = U_{t-1} \cup P$ is disjoint and covers

$$\text{Cov}(U_t) = \text{Cov}(U_{t-1}) \cup Q(q) = \bigcup_{j=1}^t Q(p(C_j)).$$

The volume update is exact by disjointness, so $y_t = 2^n - |\text{Cov}(U_t)|$ equals the number of satisfying assignments of the prefix CNF. Finally, F is unsatisfiable iff $y_m = 0$, and if $y_m > 0$ then any extracted $x \notin \text{Cov}(U_m)$ satisfies all clauses. $\square$

4.1 Witness extraction

If $y_m > 0$, one may extract a witness by finding any point outside $\text{Cov}(U_m)$. A constructive approach is a greedy coordinate-by-coordinate procedure that maintains the exact number of covered points surviving under the current partial assignment.

Proposition 4.2 (Witness extraction is polynomial in the representation size). *Given access to the pattern (bitmask) representation of a disjoint family $U \subseteq \{0, 1, \bullet\}^n$ with $\text{Cov}(U) \neq \Omega_n$, one can compute $x \in \Omega_n \setminus \text{Cov}(U)$ in time $O(n \cdot |U|)$ by a greedy procedure that maintains exact covered-point counts.*

Proof. For a partial assignment ρ , let $c(\rho) = |\text{Cov}(U) \cap Q(\rho)|$. Each cube $u \in U$ contributes $2^{n-k(u)-t}$ to $c(\rho)$, where t is the number of coordinates fixed by ρ that are free in u , provided u agrees with ρ on its fixed coordinates, and 0 otherwise; hence $c(\rho)$ is computable in time $O(|U|)$ from the bitmasks. Start with ρ empty, so $c(\rho) = |\text{Cov}(U)| < 2^n$. For $i = 1, \dots, n$: compute $c_0 = c(\rho \cup \{x_i = 0\})$ and $c_1 = c(\rho \cup \{x_i = 1\})$. The two subcubes partition $Q(\rho)$, so $c_0 + c_1 = c(\rho) < 2^{n-i+1}$, hence $\min(c_0, c_1) < 2^{n-i}$; fix x_i to the value attaining the minimum, preserving the invariant $c(\rho) < 2^{n-i}$. After n steps ρ is a complete assignment with $c(\rho) < 1$, i.e., $c(\rho) = 0$, so $\rho \notin \text{Cov}(U)$. Each of the $2n$ counts costs $O(|U|)$, for $O(n \cdot |U|)$ in total. $\square$

Remark 4.3 (Representation access is necessary). Access to the representation is essential: an algorithm limited to membership queries (“is x covered?”) may need $\Omega(2^n)$ queries in the worst case. Indeed, fix $x^* \in \Omega_n$ and let $U = \{\{x : x_1 = x_1^*, \dots, x_{i-1} = x_{i-1}^*, x_i = 1 - x_i^*\} : i \in [n]\}$. Then U is disjoint and $\text{Cov}(U) = \Omega_n \setminus \{x^*\}$: every $x \neq x^*$ disagrees with x^* at some first index i , hence lies in the i -th cube. A query transcript of length $q < 2^n$ leaves some point y unqueried and is consistent with the family built from y in place of x^* , so no such algorithm can identify the unique uncovered point.

5 Complexity in terms of fragmentation

The running time of COVERTRACE-SAT is governed by how large U becomes.

Definition 5.1 (Fragmentation parameters). Let U_t be the disjoint family after t insertions. Define the fragmentation $S_t = |U_t|$ and $S = \max_t S_t$. Let T be the total number of cube pieces produced by all calls to CubeDiff across the run (counting multiplicity).

Proposition 5.2 (Parameterized time and space bounds). *The core algorithm runs in time $O((T + m) \cdot c(n))$ and uses space $O(S \cdot c(n))$, where $c(n)$ is the cost of primitive cube operations under the chosen representation (e.g., $c(n) = O(1)$ with bitmasks plus occasional $O(n)$ scans). In the worst case, T, S can be $2^{\Theta(n)}$.*

Proof sketch. Each produced cube piece is generated, stored, and later processed by primitive operations a constant number of times, so $O(T \cdot c(n))$ accounts for all work that emits a piece. The remaining work consists of AddCube calls that emit no new piece: such a call still scans U once, and it is charged either to its seed cube q or to a piece that is destroyed by a subsequent CubeDiff and never re-created, each at most once; since there are m seeds, this adds $O(m \cdot c(n))$. The $+m$ term is necessary: with m identical clauses, $T = 0$ yet the run performs $\Theta(m)$ cube operations. Space is dominated by the maximum family size. Exponentiality follows from the lower bound in Section 6. $\square$

6 A tight exponential worst-case lower bound

We give an unconditional worst-case lower bound for any disjoint axis-aligned subcube cover of an explicit set, implying that COVERTRACE-SAT must use exponential space/time on some instances.

6.1 Parity as an extremal obstruction

For $x \in \Omega_n$, define parity $\text{par}(x) = x_1 \oplus \dots \oplus x_n$ and let $O_n = \{x : \text{par}(x) = 1\}$.

Lemma 6.1 (No nontrivial axis-aligned cube is monochromatic for parity). *If $p \in \{0, 1, \bullet\}^n$ has at least one free coordinate ($k(p) < n$), then $Q(p)$ contains both even- and odd-parity points.*

Proof. Choose a free coordinate i of p and any $x \in Q(p)$. Let x' be x with bit i flipped. Then $x' \in Q(p)$ but $\text{par}(x') = \text{par}(x) \oplus 1$. $\square$

Theorem 6.2 (Tight DSOP lower bound for parity). *Any disjoint family $U \subseteq \{0, 1, \bullet\}^n$ with $\text{Cov}(U) = O_n$ must satisfy $|U| = 2^{n-1}$, and every $u \in U$ is a singleton point ($k(u) = n$).*

Proof. By the lemma, no cube with $k(u) < n$ can be contained in O_n . Hence every $u \in U$ must be a singleton. Since the family is disjoint and covers O_n , it must contain exactly one cube for each of the $|O_n| = 2^{n-1}$ points. $\square$

6.2 A CNF family forcing exponential fragmentation

Corollary 6.3 (Worst-case fragmentation for COVERTRACE-SAT). *For each $n \geq 1$, there exists a satisfiable CNF F_n on n variables such that any disjoint subcube representation of $U(F_n)$ requires 2^{n-1} cubes. Consequently, any run of COVERTRACE-SAT on F_n must use $\Omega(2^{n-1})$ space and $\Omega(2^{n-1})$ time.*

Proof. For each $a \in O_n$, define a clause C_a falsified only by assignment a :

$$C_a = \bigvee_{i:a_i=0} x_i \vee \bigvee_{i:a_i=1} \neg x_i.$$

Then $Q(p(C_a)) = \{a\}$ and thus $U(F_n) = \bigcup_{a \in O_n} \{a\} = O_n$ for $F_n = \bigwedge_{a \in O_n} C_a$. Since $E_n = \Omega_n \setminus O_n$ is nonempty, F_n is satisfiable. Theorem 6.2 implies any disjoint subcube cover of $U(F_n)$ has size 2^{n-1} . $\square$

Remark 6.4 (Input-size caveat and a stronger open problem). The family above has 2^{n-1} clauses, so its input size is exponential in n . A substantially stronger goal is to exhibit polynomial-size CNFs F_n such that every DSOP for $\neg F_n$ has size $2^{\Omega(n)}$. This becomes an explicit lower-bound problem in knowledge compilation / proof complexity (Section 7 and Section 9).

7 COVERTRACE as knowledge compilation

7.1 DSOP / deterministic DNF

Definition 7.1 (Cubes as terms, DSOP). A pattern $p \in \{0, 1, \bullet\}^n$ corresponds to a conjunction (term)

$$T(p) = \bigwedge_{i \in \text{supp}(p)} \begin{cases} x_i & \text{if } p_i = 1, \\ \neg x_i & \text{if } p_i = 0. \end{cases}$$

A disjoint sum of products (DSOP) for a Boolean function g is a DNF $g = \bigvee_{j=1}^s T(p_j)$ such that the sets $Q(p_j)$ are pairwise disjoint. Equivalently, it is a deterministic DNF where no assignment satisfies two distinct terms.

Proposition 7.2 (Forbidden cubes are the terms of $\neg F$). For a CNF $F = \bigwedge_{j=1}^m C_j$, one has

$$\neg F \equiv \bigvee_{j=1}^m T(p(C_j)).$$

Thus $U(F)$ is the set of satisfying assignments of $\neg F$.

Theorem 7.3 (COVERTRACE compiles $\neg F$ into a DSOP). The output family U of COVERTRACE-SAT yields a DSOP for $\neg F$:

$$\neg F \equiv \bigvee_{u \in U} T(u),$$

and the terms are mutually disjoint.

Proof. By correctness, $\text{Cov}(U) = U(F)$. Each cube u corresponds to term $T(u)$ whose satisfying assignments are exactly $Q(u)$. Disjointness implies determinism. $\square$

7.2 Position in the Darwiche–Marquis map

The knowledge compilation map organizes target languages by (i) succinctness and (ii) polytime support for queries/transformations such as model counting (CT), satisfiability (SAT), clausal entailment (CE), and conditioning (CD), among others. Deterministic DNF / DSOP supports model counting in time linear in the representation size by additivity. However, deterministic DNF is generally less succinct than languages such as d-DNNF, OBDD, or SDD which also support polytime counting under additional structural constraints.

7.3 A conditional collapse of PH from uniform DSOP compilation

We now formalize a complexity-theoretic consequence suggested by the COVERTRACE viewpoint.

Theorem 7.4 (Conditional PH collapse under uniform disjoint-subcube compilation). Assume there exists a deterministic algorithm $\mathcal{A}$ such that, for every CNF formula F on n variables with size $m = |F|$, the algorithm runs in time $\text{poly}(n, m)$ and outputs a disjoint family $U \subseteq \{0, 1, \bullet\}^n$ satisfying:

1. $|U| \leq \text{poly}(n, m)$ and the encoding length of U is $\text{poly}(n, m)$;

2. $\text{Cov}(U) = U(F)$ (exact forbidden region).

Then $\#\text{SAT} \in \text{P}$ and the Polynomial-Time Hierarchy collapses: $\text{PH} = \text{P}$.

Proof. Given F , run $\mathcal{A}(F)$ to obtain U in time $\text{poly}(n, m)$. By disjointness,

$$|U(F)| = \sum_{u \in U} 2^{n-k(u)},$$

computable in polynomial time. Hence $\#\text{SAT}(F) = 2^n - |U(F)|$ is computable in polynomial time, so $\#\text{SAT} \in \text{P}$. Since $\#\text{SAT}$ is $\#\text{P}$ -complete, this implies $\#\text{P} \subseteq \text{FP}$. By Toda's theorem, $\text{PH} \subseteq \text{P}^{\#\text{P}}$; substituting the oracle with a polynomial-time algorithm yields $\text{PH} \subseteq \text{P}$. Trivially $\text{P} \subseteq \text{PH}$, hence $\text{PH} = \text{P}$. $\square$

Remark 7.5 (Why this does not prove $\text{P} \neq \text{NP}$). Theorem 7.4 is conditional and rules out a strong uniform DSOP compilation hypothesis under standard beliefs. Exponential lower bounds for DSOP (or for COVERTRACE's fragmentation) only show that this compilation language is not universally succinct; they do not rule out other algorithms for SAT.

7.4 Disjoint covers as proof certificates (UNSAT)

A disjoint family U with $\text{Cov}(U) = \Omega_n$ together with, for each $u \in U$, a witness clause $C_{j(u)} \in F$ such that $Q(u) \subseteq Q(p(C_{j(u)}))$, constitutes a certificate that F is unsatisfiable. It is verifiable in time polynomial in $|U|$, n , and m : check pairwise disjointness of the cubes, per-cube containment in the forbidden cube of its witness clause, and total volume $\sum_{u \in U} 2^{n-k(u)} = 2^n$. The containment check is essential for soundness: disjointness and total volume alone certify only that U covers Ω_n , not that the cubes are forbidden by F (for satisfiable $F = (x_1)$ on $n = 1$, the family $\{\{x_1 = 0\}, \{x_1 = 1\}\}$ would otherwise be wrongly accepted). COVERTRACE's output satisfies the containment property by construction (Lemma 2.2 in Part II). This induces a Cook–Reckhow proof system whose proofs are disjoint covers with clause witnesses; if all UNSAT instances had polynomial-size proofs in this system, then $\text{UNSAT} \in \text{NP}$ and $\text{NP} = \text{coNP}$.

8 An affine extension: exponential compression for parity

Parity is a worst-case obstruction for axis-aligned cubes, but it is a single linear equation over $\mathbb{F}_2$. This motivates extending the representation language.

8.1 Affine sets over $\mathbb{F}_2$

Definition 8.1 (Affine subspaces and cosets). Let $A \in \mathbb{F}_2^{r \times n}$ and $b \in \mathbb{F}_2^r$. The solution set

$$\mathcal{A}(A, b) = \{x \in \Omega_n : Ax = b\}$$

is an affine subspace (a coset of $\ker A$). Its size is either 0 (inconsistent) or $2^{n-\text{rank}(A)}$.

Proposition 8.2 (Counting affine solutions is polynomial). *Given A, b , one can decide emptiness and compute $|\mathcal{A}(A, b)|$ in time $O(rn^2)$ (equivalently $O(n^3)$ when $r = O(n)$) by Gaussian elimination over $\mathbb{F}_2$.*

8.2 Exponential separation: parity

Theorem 8.3 (Axis-aligned DSOP vs. affine representation for parity). *The odd-parity set O_n equals the affine set $\{x : x_1 \oplus \dots \oplus x_n = 1\}$, representable with one linear equation. However, any disjoint axis-aligned subcube cover of O_n requires 2^{n-1} cubes (Theorem 6.2). Thus allowing affine pieces yields an exponential compression for this family.*

8.3 Toward COVERTRACE-Affine

The preceding theorem suggests a principled extension: augment the disjoint-subcube language with affine pieces (or affine decision nodes), so that XOR structure can be represented compactly while retaining polytime model counting on the compiled representation. Affine compilation languages and extended affine decision trees provide relevant precedents.

9 Toward an obstruction program: fragmentation lower bounds from structure

9.1 The minimal disjoint-subcube cover number

Definition 9.1 (Disjoint subcube cover number). For a set $S \subseteq \Omega_n$, define

$$\chi_{\sqcup}(S) = \min \left\{ |U| : U \subseteq \{0, 1, \bullet\}^n \text{ disjoint and } \text{Cov}(U) = S \right\}.$$

Equivalently, $\chi_{\sqcup}(S)$ is the minimum DSOP size of the indicator function $\mathbf{1}_S$. For a CNF F , the minimal representation size of the forbidden region is $\chi_{\sqcup}(U(F))$. Any exact COVERTRACE-style algorithm that stores a disjoint cover of $U(F)$ must use at least $\chi_{\sqcup}(U(F))$ pieces in the worst case (even with optimal clause ordering and maximal local merges).

Lemma 9.2 (Trivial volumetric lower bound). *For any nonempty $S \subseteq \Omega_n$ (for $S = \emptyset$, $\chi_{\sqcup}(S) = 0$ by convention),*

$$\chi_{\sqcup}(S) \geq \frac{|S|}{\max\{|Q(p)| : p \in \{0, 1, \bullet\}^n, Q(p) \subseteq S\}}.$$

Proof. In any disjoint cover of S by cubes, each cube has size at most the stated maximum. Since the cubes are disjoint and their sizes sum to $|S|$, the number of cubes is at least the ratio. $\square$

The parity lower bound is an extremal case where the largest monochromatic cube inside O_n has size 1.

9.2 Influence as a fragmentation obstruction

For a Boolean function $f : \Omega_n \rightarrow \{0, 1\}$, the (total) influence measures the expected boundary size of its level sets.

Definition 9.3 (Influence). For $i \in [n]$, let $x^{\oplus i}$ denote x with bit i flipped. The influence of coordinate i on f is

$$\text{Inf}_i(f) = \Pr_{x \sim \text{Unif}(\Omega_n)} [f(x) \neq f(x^{\oplus i})].$$

The total influence is $\text{Inf}(f) = \sum_{i=1}^n \text{Inf}_i(f)$.

Total influence equals the normalized edge boundary of $S = f^{-1}(1)$: if $\partial_i S$ denotes the set of edges in direction i crossing between S and its complement, then $|\partial_i S| = 2^{n-1} \text{Inf}_i(f)$ and $|\partial S| = \sum_i |\partial_i S| = 2^{n-1} \text{Inf}(f)$.

Proposition 9.4 (Influence upper bound from a DSOP). *Let $S \subseteq \Omega_n$ and $f = \mathbf{1}_S$. If $S = \bigsqcup_{u \in U} Q(u)$ is a disjoint union of cubes, then*

$$\text{Inf}(f) \leq \sum_{u \in U} k(u) 2^{1-k(u)}.$$

Proof. For a cube $Q(u)$ of width $k = k(u)$, along each fixed coordinate $i \in \text{supp}(u)$ every point in $Q(u)$ has a unique neighbor outside $Q(u)$, contributing $|Q(u)| = 2^{n-k}$ crossing edges in direction i . Thus $Q(u)$ alone would contribute $\text{Inf}_i(\mathbf{1}_{Q(u)}) = 2^{n-k}/2^{n-1} = 2^{1-k}$ for each $i \in \text{supp}(u)$ and 0 otherwise, giving total influence $k2^{1-k}$. For a disjoint union, the boundary of S is a subset of the union of individual cube boundaries (adjacent cubes can only cancel boundary edges), hence the inequality after summing over cubes. $\square$

Corollary 9.5 (A width-aware influence lower bound on $\chi_\sqcup$). *Let $S \subseteq \Omega_n$, let $k_0 \geq 1$, and suppose every cube contained in S has width at least k_0 (equivalently, S contains no axis-aligned cube of dimension $> n - k_0$). Then*

$$\chi_\sqcup(S) \geq \frac{\text{Inf}(\mathbf{1}_S)}{k_0 2^{1-k_0}}.$$

Proof. In any disjoint cover of S , each cube has width $k(u) \geq k_0 \geq 1$, and the map $k \mapsto k 2^{1-k}$ is non-increasing for $k \geq 1$, hence $k(u)2^{1-k(u)} \leq k_0 2^{1-k_0}$. Apply Proposition 9.4 and rearrange. $\square$

For parity, $k_0 = n$ and $\text{Inf}(\mathbf{1}_{O_n}) = n$, so Corollary 9.5 yields $\chi_\sqcup(O_n) \geq 2^{n-1}$, matching Theorem 6.2.

9.3 Subcube partition complexity and recent structural lower bounds

Subcube partitions are a well-studied computation model: a subcube partition for a Boolean function f partitions Ω_n into disjoint subcubes on which f is constant, and its complexity is the number of parts. This measure is closely related to deterministic decision trees, but can be strictly smaller, and separations have been proved. Recent work relates subcube partition complexity to additive structure and influence of the support of f . Such results suggest a path to a priori fragmentation lower bounds for COVERTRACE on structured formulas: if $U(F)$ has strong additive structure or large influence in an appropriate sense, then $\chi_\sqcup(U(F))$ (and hence worst-case fragmentation) must be large.

Problem 9.6 (Explicit CNF $\rightarrow$ DSOP lower bounds). Exhibit an explicit family of CNFs F_n of size $\text{poly}(n)$ such that $\chi_\sqcup(U(F_n)) = 2^{\Omega(n)}$.

Proving such a lower bound would separate polynomial-size CNF representations from polynomial-size DSOP representations for $\neg F_n$ and would provide a concrete obstruction to uniform disjoint-subcube compilation.

10 A GCT-inspired viewpoint: disjoint subcubes as restricted rank-one tensors

GCT studies lower bounds in algebraic complexity by relating explicit functions to geometric and representation-theoretic obstructions. While COVERTRACE operates in a discrete setting, it induces a natural tensor decomposition viewpoint.

10.1 Truth tables as tensors

Identify a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ with its order- n tensor $T_f \in \{0, 1\}^{2 \times \dots \times 2}$, where $(T_f)_{x_1, \dots, x_n} = f(x_1, \dots, x_n)$. Let $e_0 = (1, 0)$, $e_1 = (0, 1)$, and $e_\star = (1, 1)$ in $\mathbb{R}^2$. A pattern $p \in \{0, 1, \bullet\}^n$ induces a rank-one tensor

$$T_p = v_1 \otimes \dots \otimes v_n, \quad v_i = \begin{cases} e_0 & p_i = 0, \\ e_1 & p_i = 1, \\ e_\star & p_i = \bullet. \end{cases}$$

One checks that T_p is the indicator tensor of the cube $Q(p)$.

Proposition 10.1 (DSOP as restricted nonnegative tensor rank). *Let $S \subseteq \Omega_n$ and $f = \mathbf{1}_S$. Then $\chi_{\sqcup}(S)$ equals the minimum r such that*

$$T_f = \sum_{j=1}^r T_{p_j}$$

with patterns $p_j \in \{0, 1, \bullet\}^n$ whose cubes are pairwise disjoint.

Remark 10.2. Ignoring the disjointness constraint yields a restricted nonnegative tensor rank measure (since all tensors have nonnegative entries). Allowing cancellations (signed sums) corresponds to inclusion–exclusion and can dramatically reduce representation size (as for parity), paralleling the gap between explicit sums and compact arithmetic circuits.

10.2 Obstructions and open problems

This tensor viewpoint suggests a discrete “GCT-like” program: identify invariants of $T_{\mathbf{1}_{U(F)}}$ (under the natural action of the hypercube automorphism group) that lower bound $\chi_{\sqcup}(U(F))$. Even partial progress—e.g., proving superpolynomial $\chi_{\sqcup}$ for an explicit polynomial-size CNF family—would be a meaningful lower bound in a concrete representation model.

11 Conclusion

COVERTRACE-SAT gives a simple and exact geometric compilation of $\neg F$ into a disjoint subcube cover, enabling exact #SAT and witness extraction in time polynomial in the representation size whenever fragmentation remains controlled. We proved a tight exponential worst-case lower bound using parity, and we formalized a strong conditional consequence: a uniform polynomial-time disjoint-subcube compiler for all CNFs would collapse PH. We proposed an affine extension that compresses parity exponentially and articulated GCT-inspired obstructions via a restricted tensor-rank interpretation.

The remaining frontier is explicit lower bounds for polynomial-size CNFs in the DSOP language and principled hybrid compilation strategies that combine axis-aligned and affine structure while retaining exactness.

A Implementation notes and heuristics (not part of the core mathematics)

This appendix collects optimizations that can significantly improve practical performance but do not change the formal core (Algorithms 1–3) and do not evade worst-case lower bounds.

A.1 Bitmask representation

A pattern $p \in \{0, 1, \bullet\}^n$ can be stored as two n -bit masks: a fixed-mask F indicating which coordinates are fixed and a value-mask V specifying their values. Intersection and containment can then be tested with constant-time bit operations.

A.2 Buddy merges

If two cubes have the same fixed coordinates and differ in exactly one fixed bit, they are “buddies” and can be merged by freeing that bit, reducing fragmentation without changing the covered set. Repeated merges correspond to local boundary cancellation.

A.3 Clause ordering

Processing clauses that forbid large-volume cubes early (e.g., short clauses) may reduce intermediate fragmentation on some instances. Such orderings are instance-dependent heuristics and are excluded from the core correctness analysis.

B A quantum-computing viewpoint (optional)

The hypercube Ω_n indexes the computational basis $\{|x\rangle : x \in \{0, 1\}^n\}$ of n qubits. A cube $Q(p)$ corresponds to the span of basis states consistent with p ; disjointness of cubes corresponds to orthogonality of these subspaces. Exact model counting $\#\text{SAT}(F)$ equals the dimension of the satisfying subspace.

While Grover’s algorithm provides a quadratic speedup for searching satisfying assignments given an oracle, it does not yield polynomial-time SAT in general. Nevertheless, COVERTRACE-style geometric structure could inform oracle construction or hybrid classical-quantum heuristics (e.g., amplitude amplification restricted to surviving regions), although this remains speculative.

References (Part I)

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Parte II

Extensiones al Marco de Compilación por Subcubos Disjuntos: Fragmentación, Resolución Arbórea, Espectro de Fourier y Representaciones Poliédricas

Extensiones al Marco de Compilación por Subcubos Disjuntos

Fragmentación, Resolución Arbórea, Espectro de Fourier y Representaciones Poliédricas

Oscar Riveros

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Versión corregida conforme al informe de revisión de pares del 14 de agosto de 2026: el teorema de equivalencia partición–resolución arbórea se reemplaza por la desigualdad válida con contraejemplo explícito y árbol refutante de 7 hojas verificado; el Corolario sobre PHP se degrada a problema abierto (la cota volumétrica da solo la constante 4); la inmersión TCOC se reformula como identidad de codificación exacta sin reclamar la Definición 4.1 externa, y la afirmación de NP-completitud se reemplaza por enunciados de complejidad precisos; correcciones en la identidad influencia–Fourier (factor 4), en la prueba del Lema invariante y en las citas al corpus.

Resumen

Se extiende el marco COVERTRACE-SAT de las *Obras Completas* con tres resultados nuevos. (1) **Partición de búsqueda y resolución arbórea**: $\mathcal{S}_{\text{part}}(F) \leq \mathcal{S}_{\text{tr}}(F)$ para toda CNF insatisfacible, y la igualdad es falsa en general: se exhibe una CNF insatisfacible explícita sobre $n = 4$ variables con $\mathcal{S}_{\text{part}}(F) = 6 < 7 = \mathcal{S}_{\text{tr}}(F)$, con árbol refutante de 7 hojas verificado (Proposición 2.4). Para COVERTRACE sobre PHP_n^{n+1} , la cota volumétrica directa da solo la constante 4 (las cláusulas de hoyo tienen ancho 2): la fragmentación exponencial en PHP y la transferencia de cotas de resolución arbórea para Tseitin quedan como problemas abiertos (Corolario 2.6). (2) **Cota espectral de Fourier–Walsh**: $\chi_{\sqcup}(S) \geq 2^d \cdot \max_{|\alpha|=d} |\widehat{\mathbf{1}_S}(\alpha)|$, generalizando la Proposición 9.4 de la Parte I, ajustada para paridad (2^{n-1}) y con cota volumétrica complementaria para mayoría ($2^{\Omega(n)}$). (3) **Certificado poliédrico y codificación exacta**: cada cobertura de $U(F)$ induce K desigualdades lineales que describen $\text{Mod}(F)$ módulo integralidad, y la codificación $\phi_n(s)_j = [s \in Q(u_j)]$ da la identidad exacta $\kappa_n(s) = [s \in U(F)]$, con $|U| = \chi_{\sqcup}(U(F))$ cubos en el caso mínimo; la decisión asociada es polinomial con entrada explícita y equivalente a SAT con entrada CNF, sin reducción de Karp. El Problema 9.6 de la Parte I (exhibir CNFs polinomiales con $\chi_{\sqcup}$ exponencial) se delimita como la separación CNF vs. DSOP en el mapa de compilación de conocimiento.

1 Preliminares

1.1 Hipercubo booleano y subcubos

Sea $\Omega_n = \{0, 1\}^n$. Un **patrón** es $p \in \{0, 1, \bullet\}^n$. El **subcubo inducido** es

$$Q(p) = \{x \in \Omega_n : \forall i \in [n], p_i \neq \bullet \Rightarrow x_i = p_i\}. \quad (1)$$

$\text{supp}(p) = \{i : p_i \in \{0, 1\}\}$ (fijas), ancho $k(p) = |\text{supp}(p)|$, volumen $\text{vol}(p) = |Q(p)| = 2^{n-k(p)}$. Una familia $U \subseteq \{0, 1, \bullet\}^n$ es **disjunta** si $Q(u) \cap Q(v) = \emptyset$ para $u \neq v$, con $\text{Cov}(U) = \bigcup_{u \in U} Q(u)$.

1.2 Codificación SAT

Para una cláusula $C = \bigvee_{i \in P^+} x_i \vee \bigvee_{i \in P^-} \neg x_i$, el **patrón de cláusula** es $p(C)_i = 0$ si $x_i \in C$, 1 si $\neg x_i \in C$, $\bullet$ en otro caso. La **región prohibida** de $F = \bigwedge_{j=1}^m C_j$ es

$$U(F) = \bigcup_{j=1}^m Q(p(C_j)) \subseteq \Omega_n, \quad (2)$$

con $\#\text{SAT}(F) = 2^n - |U(F)|$.

1.3 Número de cobertura disjunta

Para $S \subseteq \Omega_n$,

$$\chi_{\sqcup}(S) = \min\{|U| : U \text{ disjunta, } \text{Cov}(U) = S\}, \quad (3)$$

el número mínimo de términos en un DSOP de $\mathbf{1}_S$.

1.4 COVERTRACE-SAT y fragmentación

El algoritmo mantiene U_t disjunta con $\text{Cov}(U_t) = \bigcup_{j=1}^t Q(p(C_j))$. En cada paso, **CubeDiff** (Parte I, Algoritmo 1) inserta $Q(p(C_{t+1})) \setminus \text{Cov}(U_t)$. La fragmentación es $S_t = |U_t|$, con pico $S_\pi = \max_t S_t$ y mínima incremental

$$S_{\text{inc}}(F) = \min_{\pi \in \mathfrak{S}_m} \max_{t=1}^m |U_t^{(\pi)}| \geq \chi_{\sqcup}(U(F)). \quad (4)$$

1.5 Resultados de la Parte I utilizados

- **Parte I, Teo. 4.1:** invariante de COVERTRACE. **Parte I, Lema 3.2:** $\text{CubeDiff}(p, r)$ produce $\leq |\text{supp}(r) \setminus \text{supp}(p)|$ subcubos.
- **Parte I, Teo. 6.2:** $\chi_{\sqcup}(O_n) = 2^{n-1}$ (O_n paridad impar). **Parte I, Teo. 7.4:** compilación DSOP uniforme polinomial $\Rightarrow \text{PH} = \text{P}$.
- **Parte I, Prop. 9.4:** $\text{Inf}(\mathbf{1}_S) \leq \sum_{u \in U} k(u) 2^{1-k(u)}$. **Parte I, Lema 9.2:** $\chi_{\sqcup}(S) \geq |S| / \max_{Q \subseteq S} \text{vol}(Q)$.
- **TCOC** (documento externo, citado solo como trasfondo): Definición 4.1 (representación exacta de optimización, con ϕ_n biyectiva y ψ_n inversa), §7 (barrera de materialización) y Teo. 9.5 (seis eslabones para $\text{P} = \text{NP}$). Ningún resultado de esta Parte invoca esas nociones.

2 Particiones de búsqueda y resolución arbórea: desigualdad, contraejemplo y fragmentación

Sea $F = C_1 \wedge \dots \wedge C_m$ una CNF **insatisfacible**. Al finalizar COVERTRACE, $\text{Cov}(U_m) = U(F) = \Omega_n$: U_m particiona el hipercubo en subcubos disjuntos.

Definición 2.1 (Partición de búsqueda). $U = \{u_1, \dots, u_K\}$ es una **partición de búsqueda** para F si $\text{Cov}(U) = \Omega_n$ (disjunta) y $\forall j \exists C_{i_j} \in F : Q(u_j) \subseteq Q(p(C_{i_j}))$.

Lema 2.2 (Invariante de contención). Sea U_m la familia producida por COVERTRACE-SAT con orden π . Para cada $u \in U_m$ existe $j \in [m]$ con $Q(u) \subseteq Q(p(C_j))$.

Demostración. Inducción en t . En el paso $t + 1$, AddCube (Parte I, Algoritmo 2) parte de $P = \{Q(p(C_{t+1}))\}$ y ejecuta CubeDiff(p, R) con p pieza candidata y $R \in U_t$; cada llamada devuelve subcubos de $Q(p)$ (CubeDiff solo fija coordenadas libres del primer argumento). Por tanto $U_{t+1} = U_t \cup P$ con $P \subseteq Q(p(C_{t+1}))$: las piezas viejas conservan su cláusula testigo C_{j_v} por hipótesis inductiva, y cada pieza nueva hereda C_{t+1} . $\square$

Definanse

$$\mathcal{S}_{\text{part}}(F) = \min\{|U| : U \text{ partición de búsqueda}\},$$

y

$$\mathcal{S}_{\text{tr}}(F) = \min\{\# \text{ hojas de refutación arbórea de } F\}.$$

Teorema 2.3 (Desigualdad partición–resolución arbórea). Para toda CNF insatisfacible F , $\mathcal{S}_{\text{part}}(F) \leq \mathcal{S}_{\text{tr}}(F)$.

Demostración. Sea Π una refutación arbórea con S hojas. Árbol de búsqueda dual: en la raíz ($\perp$), para $C = \text{Res}(C_L, C_R, x_i)$ divídase sobre x_i : $x_i = 0 \rightarrow C_L$, $x_i = 1 \rightarrow C_R$. Invariante inductivo: la asignación parcial ρ del camino falsifica la cláusula del nodo. En la raíz, $\emptyset$ falsifica $\perp$. Si ρ falsifica C , todo literal de C_L es x_i o pertenece a C (pues $C = (C_L \cup C_R) \setminus \{x_i, \neg x_i\}$): con $x_i = 0$ se falsifica x_i y ρ falsifica el resto, luego $\rho \cup \{x_i = 0\}$ falsifica C_L ; análogamente $\rho \cup \{x_i = 1\}$ falsifica C_R (que contiene $\neg x_i$). Cada hoja es un axioma $C \in F$ y su camino ρ lo falsifica. Las S hojas definen subcubos disjuntos (dos hojas divergen en alguna variable consultada con valores distintos) que particionan Ω_n (toda asignación completa sigue exactamente un camino), cada uno $\subseteq Q(p(C))$. Por tanto $\mathcal{S}_{\text{part}}(F) \leq S$. $\square$

La dirección recíproca es **falsa**: una partición de búsqueda con K piezas no implica una refutación arbórea con K hojas. La hoja de un árbol de decisión refutante es válida si y solo si su asignación parcial falsifica una cláusula; el árbol que “localiza” la pieza u_j debe consultar todas las variables de $\text{supp}(u_j)$, y las ramas de descarte (cuando la variable consultada no coincide con $p(u_j)$) producen asignaciones parciales que en general no falsifican cláusula alguna, obligando a hojas adicionales.

Proposición 2.4 (La igualdad $\mathcal{S}_{\text{part}} = \mathcal{S}_{\text{tr}}$ es falsa). Existe una CNF insatisfacible F sobre $n = 4$ variables con $\mathcal{S}_{\text{part}}(F) = 6 < 7 = \mathcal{S}_{\text{tr}}(F)$.

Demostración. Considérese la CNF (insatisfacible por búsqueda exhaustiva sobre $\{0, 1\}^4$)

$$\begin{aligned} F = & (x_0 \vee \neg x_1 \vee x_2) \wedge (x_2 \vee x_3) \wedge (x_0 \vee x_3) \wedge (x_1 \vee x_3) \\ & \wedge (x_0 \vee x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_0 \vee \neg x_1 \vee \neg x_3) \wedge (x_0 \vee \neg x_2) \\ & \wedge (\neg x_0 \vee x_1 \vee \neg x_2) \wedge (x_0 \vee \neg x_2 \vee x_3) \wedge (x_3) \\ & \wedge (\neg x_0 \vee \neg x_1 \vee x_2) \wedge (x_0 \vee \neg x_1 \vee \neg x_2 \vee \neg x_3) \wedge (x_1 \vee x_2 \vee \neg x_3). \end{aligned}$$

Cota superior $\mathcal{S}_{\text{part}}(F) \leq 6$. La siguiente familia es una partición de búsqueda (patrones $x_0 x_1 x_2 x_3$ con $\bullet$ = libre; la última columna da la cláusula testigo con $Q(u) \subseteq Q(p(C))$):

u	$\text{vol}(u)$	C testigo	verificación
$\bullet \bullet \bullet 0$	8	(x_3)	$x_3 = 0$ falsifica (x_3)
$\bullet 0 0 1$	2	$(x_1 \vee x_2 \vee \neg x_3)$	$x_1 = x_2 = 0, x_3 = 1$ la falsifica
$0 \bullet 1 1$	2	$(x_0 \vee \neg x_2)$	$x_0 = 0, x_2 = 1$ la falsifica
$0 1 0 1$	1	$(x_0 \vee \neg x_1 \vee x_2)$	$x_0 = x_2 = 0, x_1 = 1$ la falsifica
$1 0 1 1$	1	$(\neg x_0 \vee x_1 \vee \neg x_2)$	$x_0 = 1, x_1 = 0, x_2 = 1$ la falsifica
$1 1 \bullet 1$	2	$(\neg x_0 \vee \neg x_1 \vee \neg x_3)$	$x_0 = x_1 = 1, x_3 = 1$ la falsifica

Los volúmenes suman $8 + 2 + 2 + 1 + 1 + 2 = 16$: la familia es disjunta y cubre $\{0, 1\}^4$.

Cotas inferiores y $\mathcal{S}_{\text{tr}}(F) = 7$. Se verificaron con procedimientos exactos e independientes: (i) programación dinámica de recubrimiento mínimo por subcubos permitidos, y una búsqueda recursiva independiente de recubrimiento mínimo, confirman que no existe partición de búsqueda con ≤ 5 piezas: $\mathcal{S}_{\text{part}}(F) = 6$; (ii) el punto fijo exacto $D(C) =$ mínimo número de hojas de una derivación arbórea por resolución de la cláusula C a partir de los axiomas F da $D(\perp) = 7$: $\mathcal{S}_{\text{tr}}(F) = 7$. Este valor se corrobora por la equivalencia clásica entre refutaciones de resolución arbórea con L hojas y árboles de decisión refutantes con L hojas (ver, p. ej., Beame–Karp–Pitassi–Saks, 2002): la enumeración exhaustiva de todos los árboles de decisión sobre 4 variables con ≤ 6 hojas no produce ninguno refutante, mientras que el siguiente árbol con 7 hojas sí es refutante (cada hoja lista la cláusula que su asignación parcial falsifica):

Consultar x_3 :

H1. $x_3 = 0$: **hoja** que falsifica (x_3) .

H2. $x_3 = 1$: consultar x_0 :

H2.1. $x_0 = 0$: consultar x_2 :

H2.1.1. $x_2 = 1$: **hoja** que falsifica $(x_0 \vee \neg x_2)$.

H2.1.2. $x_2 = 0$: consultar x_1 :

H2.1.2.1. $x_1 = 0$: **hoja** que falsifica $(x_1 \vee x_2 \vee \neg x_3)$.

H2.1.2.2. $x_1 = 1$: **hoja** que falsifica $(x_0 \vee \neg x_1 \vee x_2)$.

H2.2. $x_0 = 1$: consultar x_1 :

H2.2.1. $x_1 = 1$: **hoja** que falsifica $(\neg x_0 \vee \neg x_1 \vee \neg x_3)$.

H2.2.2. $x_1 = 0$: consultar x_2 :

H2.2.2.1. $x_2 = 0$: **hoja** que falsifica $(x_1 \vee x_2 \vee \neg x_3)$.

H2.2.2.2. $x_2 = 1$: **hoja** que falsifica $(\neg x_0 \vee x_1 \vee \neg x_2)$.

Cada hoja fue verificada por computación exacta: la asignación parcial del camino falsifica la cláusula indicada, los siete cubos son disjuntos y sus volúmenes suman $8 + 2 + 1 + 1 + 2 + 1 + 1 = 16$, de modo que el árbol particiona $\{0, 1\}^4$. En consecuencia $\mathcal{S}_{\text{part}}(F) = 6 < 7 = \mathcal{S}_{\text{tr}}(F)$ y la igualdad no se cumple en general. $\square$

Observación 2.5 (Consistencia con la literatura). La posibilidad $\mathcal{S}_{\text{part}} < \mathcal{S}_{\text{tr}}$ es consistente con resultados conocidos: la complejidad de partición en subcubos puede ser estrictamente menor que la complejidad de árbol de decisión (Kothari–Meka–Raghavendra, 2015). El Corolario siguiente usa solo la cadena válida $\mathcal{S}_{\text{peak}} \geq |U_m^{(\pi)}| \geq \mathcal{S}_{\text{part}}(F)$.

Corolario 2.6 (Fragmentación de COVERTRACE en instancias insatisfacibles). $\mathcal{S}_{\text{peak}}(F, \pi) \geq |U_m^{(\pi)}| \geq \mathcal{S}_{\text{part}}(F)$ (Lema 2.2). De aquí:

- **PHP:** para PHP_n^{n+1} con $N = n(n+1)$ variables y $n \geq 2$, las cláusulas de pichón $\bigvee_{k \in [n]} x_{i,k}$ (ancho n) fijan n variables a 0 (volumen 2^{N-n}), y las cláusulas de hoyo $\neg x_{i,k} \vee \neg x_{j,k}$ (ancho 2) fijan 2 variables a 1 (volumen 2^{N-2}). Toda pieza admisible tiene volumen $\leq 2^{N-2}$, y la cota volumétrica (Parte I, Lema 9.2) da solo

$$\mathcal{S}_{\text{part}}(\text{PHP}_n^{n+1}) \geq \frac{2^N}{2^{N-2}} = 4,$$

una constante. Determinar si la fragmentación de COVERTRACE sobre PHP es $2^{\Omega(N)}$ queda como problema abierto.

- **Tseitin:** las cotas exponenciales de resolución arbórea para Tseitin sobre expansores (Urquhart, 1987) **no** se transfieren a $\mathcal{S}_{\text{part}}$, porque la igualdad $\mathcal{S}_{\text{part}} = \mathcal{S}_{\text{tr}}$ es falsa (Proposición 2.4); la cota volumétrica da solo una constante (cláusulas de grado $\leq d$: volumen $\geq 2^{N-d}$). Determinar si $\mathcal{S}_{\text{part}}(\text{Tseitin}) = 2^{\Omega(N)}$ queda como problema abierto.

Observación 2.7 ($\mathcal{S}_{\text{part}}$ vs. $\chi_{\sqcup}$). $\mathcal{S}_{\text{part}}(F)$ restringe cada subcubo a estar contenido en $Q(p(C))$ para alguna cláusula. Para F insatisfacible, $\chi_{\sqcup}(\Omega_n) = 1$ pero $\mathcal{S}_{\text{part}}(F)$ puede ser estrictamente mayor (la Proposición 2.4 da $\mathcal{S}_{\text{part}}(F) = 6 > 1$). Si $\mathcal{S}_{\text{part}}(F)$ puede llegar a ser exponencial

queda sin demostrar en este trabajo: el único mecanismo de cota de esta sección (el volumétrico) da solo una constante para PHP. El Corolario 2.6 acota la fragmentación **algorítmica**, no el mínimo combinatorio $\chi_{\sqcup}$.

3 Cota espectral de Fourier–Walsh para $\chi_{\sqcup}$

3.1 Preliminares

Para $f : \Omega_n \rightarrow \mathbb{R}$, los coeficientes de Fourier–Walsh normalizados son

$$\widehat{f}(\alpha) = \frac{1}{2^n} \sum_{x \in \Omega_n} f(x) \chi_{\alpha}(x), \quad \chi_{\alpha}(x) = (-1)^{\alpha \cdot x}, \quad \alpha \subseteq [n].$$

Para funciones 0/1 (como todas las f de este documento), la identidad de influencia–Fourier es

$$\text{Inf}_i(f) = \Pr_x[f(x) \neq f(x^{\oplus i})] = 4 \sum_{\alpha \ni i} \widehat{f}(\alpha)^2,$$

pues $f(x) \neq f(x^{\oplus i})$ equivale a $g(x) \neq g(x^{\oplus i})$ para $g = 1 - 2f \in \{\pm 1\}$, y $\widehat{g}(\alpha) = -2\widehat{f}(\alpha)$ para $\alpha \ni i$. En consecuencia,

$$\text{Inf}(f) = 4 \sum_{\alpha} |\alpha| \widehat{f}(\alpha)^2.$$

(La identidad sin el factor 4 vale para funciones con valores ± 1 , no para indicadores 0/1.)

3.2 Espectro de Fourier de un subcubo

Lema 3.1. Sea $p \in \{0, 1, \bullet\}^n$, $S = \text{supp}(p)$ y $w = n - |S|$ la dimensión de $Q(p)$. Para $f = \mathbf{1}_{Q(p)}$ y $\alpha \subseteq [n]$:

$$\widehat{f}(\alpha) = \begin{cases} 2^{w-n} \cdot (-1)^{\alpha \cdot p|_{\alpha}} = 2^{-|S|} \cdot (-1)^{\alpha \cdot p|_{\alpha}}, & \alpha \subseteq S, \\ 0, & \text{en otro caso.} \end{cases}$$

En particular $|\widehat{f}(\alpha)| = 2^{-|S|}$ si $\alpha \subseteq S$, 0 en otro caso, y $\|\widehat{f}\|_1 = 1$.

Demostración. $\widehat{f}(\alpha) = \frac{1}{2^n} \sum_{x \in Q(p)} \chi_{\alpha}(x)$. Si $\alpha \not\subseteq S$, existe $i \in \alpha \setminus S$; las mitades $x_i = 0, 1$ de $Q(p)$ cancelan χ_{α} . Si $\alpha \subseteq S$, $\chi_{\alpha}(x) = (-1)^{\alpha \cdot p|_{\alpha}}$ constante sobre $Q(p)$, y la suma es $|Q(p)| \cdot (-1)^{\alpha \cdot p|_{\alpha}} = 2^w \cdot (-1)^{\alpha \cdot p|_{\alpha}}$. $\square$

3.3 Teorema espectral

Teorema 3.2 (Cota espectral para $\chi_{\sqcup}$). Para $S \subseteq \Omega_n$, $f = \mathbf{1}_S$, y todo $d \in \{0, \dots, n\}$,

$$\chi_{\sqcup}(S) \geq 2^d \cdot \max_{|\alpha|=d} |\widehat{f}(\alpha)|.$$

Demostración. Sea $U = \{u_1, \dots, u_K\}$ familia disjunta con $\text{Cov}(U) = S$, $K = \chi_{\sqcup}(S)$. Por disjunción $f = \sum_{j=1}^K \mathbf{1}_{Q(u_j)}$ puntualmente. Por linealidad y el Lema 3.1,

$$|\widehat{f}(\alpha)| \leq \sum_{j: \alpha \subseteq \text{supp}(u_j)} 2^{-|\text{supp}(u_j)|} \leq K \cdot 2^{-|\alpha|},$$

pues $|\text{supp}(u_j)| \geq |\alpha|$ y hay $\leq K$ términos. Despejando, $K \geq |\widehat{f}(\alpha)| \cdot 2^{|\alpha|}$. $\square$

3.4 Aplicaciones

Ejemplo 3.3 (Paridad impar). $S = O_n = \{x : \bigoplus_i x_i = 1\}$. $f = \frac{1}{2} - \frac{1}{2}\chi_{[n]}$; $\widehat{f}([n]) = -\frac{1}{2}$, resto nulos salvo $\emptyset$. Con $d = n$: $\chi_{\sqcup}(O_n) \geq 2^{n-1}$, ajustado (Parte I, Teo. 6.2).

Ejemplo 3.4 (Mayoría, cota volumétrica). $S = \text{MAJ}_n = \{x : \sum_i x_i > n/2\}$. El mayor subcubo en MAJ_n fija $\lfloor n/2 \rfloor + 1$ variables a 1 (toda extensión de un patrón con a unos fijos y b ceros fijos tiene peso mínimo a , y se requiere $a > n/2$); su volumen es $2^{n-\lfloor n/2 \rfloor-1} = 2^{\lfloor n/2 \rfloor-1}$. Como $|\text{MAJ}_n| = 2^{n-1} - \frac{1}{2}\binom{n}{n/2}$ para n par y 2^{n-1} para n impar —en todo caso $2^{n-1}(1 - o(1))$, pues $\binom{n}{\lfloor n/2 \rfloor} \leq 2^n / \sqrt{\pi n/2}$ —, el Lema 9.2 de la Parte I ($\chi_{\sqcup}(S) \geq |S| / \max_{Q \subseteq S} \text{vol}(Q)$) da

$$\chi_{\sqcup}(\text{MAJ}_n) \geq \frac{2^{n-1}(1 - o(1))}{2^{\lfloor n/2 \rfloor-1}} = 2^{n-\lfloor n/2 \rfloor}(1 - o(1)) = 2^{\lfloor n/2 \rfloor}(1 - o(1)) = 2^{\Omega(n)}.$$

3.5 Complementariedad con la cota de influencia

La Proposición 9.4 de la Parte I da $\text{Inf}(\mathbf{1}_S) \leq \sum_{u \in U} k(u) 2^{1-k(u)}$. La identidad $\text{Inf}(\mathbf{1}_S) = 4 \sum_{\alpha} |\alpha| \widehat{f}(\alpha)^2$ muestra (heurísticamente) que la cota espectral es más fuerte en alta frecuencia (grado alto), mientras que la de influencia domina en baja frecuencia. Ambas son complementarias.

4 Certificado poliédrico y codificación exacta

4.1 De subcubos a desigualdades lineales

Teorema 4.1 (Certificado poliédrico). Sea F una CNF y $U = \{u_1, \dots, u_K\}$ una familia (no necesariamente disjunta) con $\text{Cov}(U) = U(F)$. Para cada $u \in U$ defínase

$$g_u(x) = \sum_{i:p(u)_i=0} x_i + \sum_{i:p(u)_i=1} (1 - x_i).$$

Entonces, para todo $x \in \{0, 1\}^n$: $x \in Q(u) \iff g_u(x) = 0$, y

$$x \in \text{Mod}(F) \iff g_u(x) \geq 1 \ \forall u \in U,$$

es decir, $\text{Mod}(F) = \{x \in \{0, 1\}^n : g_u(x) \geq 1 \ \forall u \in U\}$. El sistema de K desigualdades lineales más $2n$ restricciones de caja describe $\text{Mod}(F)$ **módulo integralidad** (sus puntos enteros son exactamente $\text{Mod}(F)$); cada desigualdad tiene tamaño $O(n)$ y el sistema es computable en $O(Kn)$.

Demostración. Si $x \in Q(u) \cap \{0, 1\}^n$, $\forall i \in \text{supp}(u) : x_i = p(u)_i$, luego $g_u(x) = 0$. Si $x \in \{0, 1\}^n \setminus Q(u)$, existe $i \in \text{supp}(u)$ con $x_i \neq p(u)_i$; el término correspondiente de g_u vale 1 y todos los términos son no negativos, luego $g_u(x) \geq 1$. Por tanto $x \in \text{Mod}(F) \iff x \notin Q(u) \ \forall u \iff g_u(x) \geq 1 \ \forall u$. (Sobre $[0, 1]^n$ fraccionario el bicondicional no es válido: $g_u(x) = 0$ solo fuerza $x_i = p(u)_i$ en las coordenadas fijas; por ello el enunciado se restringe a $\{0, 1\}^n$.) $\square$

Observación 4.2 (La disyuntividad no se usa). El Teorema 4.1 usa solo la cobertura $\text{Cov}(U) = U(F)$, no la disyuntividad de U . La disyuntividad es la propiedad que hace aditiva la cuenta exacta $|U(F)| = \sum_{u \in U} \text{vol}(u)$ en COVERTRACE, pero no es necesaria para describir $\text{Mod}(F)$ módulo integralidad.

4.2 Codificación exacta del predicado SAT

En una versión anterior se afirmaba que la tupla $(Q_n, D_n, \phi_n, \psi_n, \Lambda_n, \beta_n)$ de abajo satisface la Definición 4.1 del documento TCOC citado. Esa Definición exige $\phi_n : S_n \rightarrow D_n$ **biyectiva** y $\psi_n : D_n \rightarrow S_n$ su **inversa**; la construcción siguiente colapsa todos los modelos en $\mathbf{0}$, de modo que ϕ_n no es inyectiva y ψ_n no está definida. El resultado se reformula, por tanto, como una identidad de codificación exacta, sin reclamar la Definición 4.1.

Teorema 4.3 (Codificación exacta del predicado SAT vía subcubos). *Sea $U = \{u_1, \dots, u_K\}$ una cobertura disjunta de $U(F)$. Defínase $\phi_n : \Omega_n \rightarrow \{0, 1\}^K$ por $\phi_n(s)_j = [s \in Q(u_j)]$ y $\Lambda_n = (1, \dots, 1) \in \mathbb{R}^K$. Entonces*

$$\kappa_n(s) := \Lambda_n \cdot \phi_n(s) = [s \in U(F)] = \begin{cases} 0, & s \in \text{Mod}(F), \\ 1, & s \in U(F). \end{cases}$$

Si además $|U| = \chi_\square(U(F))$, la codificación usa exactamente $\chi_\square(U(F))$ cubos.

Demostración. Como U es disjunta, $\phi_n(s)$ tiene a lo sumo una entrada igual a 1: para $s \in \text{Mod}(F)$ vale $\mathbf{0}$, y para $s \in U(F)$ vale $\mathbf{e}_j$, donde u_j es el único cubo que contiene a s . La identidad sigue directamente. $\square$

Proposición 4.4 (Complejidad de la decisión asociada). *Sean $D_n = \phi_n(\Omega_n)$ y $Q_n = \text{conv}(D_n)$. Entonces $D_n = \{\mathbf{0}\} \cup \{\mathbf{e}_1, \dots, \mathbf{e}_K\}$ (con $\mathbf{0}$ excluido si F es insatisfacible), y*

$$\min_{x \in Q_n} \Lambda_n \cdot x = 0 \iff F \text{ es satisfacible,}$$

(el mínimo es 0 si $\mathbf{0} \in D_n$ y 1 en otro caso; $\mathbf{0}$ es vértice de Q_n cuando pertenece a D_n). En cuanto a complejidad:

1. **Entrada explícita** (D_n, Λ_n) : decidir “ $\min_{x \in Q_n} \Lambda_n \cdot x < 1$ ” equivale a comprobar $\mathbf{0} \in D_n$ y es decidible en tiempo polinomial.
2. **Entrada CNF** F : decidir “ $\min < 1$ ” equivale a SAT, que es NP-completo; pero construir D_n desde F no es polinomial en general: $K = \chi_\square(U(F))$ puede ser exponencial (2^{n-1} para paridad, Parte I, Teo. 6.2) y evaluar ϕ_n sobre los 2^n puntos de Ω_n es exponencial. No se afirma reducción de Karp alguna.

Demostración. El mínimo de una función lineal sobre un politopo se alcanza en un vértice, y los vértices de Q_n están en D_n . Además $\Lambda_n \cdot \mathbf{0} = 0$ mientras $\Lambda_n \cdot \mathbf{e}_j = 1$, y $\mathbf{0}$ no puede escribirse como combinación convexa de los $\mathbf{e}_j$, luego es vértice cuando pertenece a D_n . Como $\text{Cov}(U) = U(F)$, todo u_j contiene algún punto de $U(F)$, de modo que $\mathbf{e}_j \in D_n$ para todo j . Por tanto $D_n = \{\mathbf{0}\} \cup \{\mathbf{e}_1, \dots, \mathbf{e}_K\}$ si $\text{Mod}(F) \neq \emptyset$ y $D_n = \{\mathbf{e}_1, \dots, \mathbf{e}_K\}$ si no; la equivalencia $\min = 0 \iff F$ satisfacible sigue. Los enunciados de complejidad son inmediatos de la descripción de D_n . $\square$

Observación 4.5 (La barrera de materialización no se hereda). La Proposición 4.4 **no** implica que las representaciones compactas del conjunto de modelos estén bloqueadas: las m desigualdades $g_{p(C_j)}(x) \geq 1$ (una por cláusula, sin disyuntividad alguna) más las $2n$ restricciones de caja ya describen $\text{Mod}(F)$ módulo integralidad con tamaño $O(mn)$, computables en $O(mn)$, pues $g_{p(C_j)}(x) \geq 1 \iff x$ satisface C_j sobre $\{0, 1\}^n$ (Teorema 4.1). Lo costoso no es describir $\text{Mod}(F)$ módulo integralidad, sino optimizar sobre el politopo fraccionario (la brecha de integralidad), una cuestión distinta del costo $\Omega(2^n)$ de materializar una cobertura disjunta de $U(F)$.

5 El Problema 9.6: estado y delimitación

5.1 Enunciado

Problema abierto (Problema 9.6 de la Parte I). Exhibir una familia explícita de CNFs F_n de tamaño $\text{poly}(n)$ tal que $\chi_{\sqcup}(U(F_n)) = 2^{\Omega(n)}$.

Equivale a **separar CNF de DSOP** en el mapa de compilación de conocimiento (Darwiche–Marquis, 2002): $f_n = \neg F_n$ admite CNF de tamaño polinomial pero toda DSOP de f_n requiere tamaño $2^{\Omega(n)}$.

5.2 Estado del arte

- (1) **Cotas triviales:** $\chi_{\sqcup}(U(F)) \leq 2^n$ (singletons). Para m cláusulas, $\chi_{\sqcup}(U(F)) \leq m \cdot 2^n$.
- (2) **Relación con $P \neq NP$:** la respuesta afirmativa **no implica** $P \neq NP$ (Remark 7.5 de la Parte I: cotas para un lenguaje de representación no separan clases). Recíprocamente, el Teorema 7.4 de la Parte I muestra que un compilador DSOP uniforme polinomial $\Rightarrow PH = P$.
- (3) **Herramientas:** la cota espectral (Teorema 3.2) y la cota volumétrica (Parte I, Lema 9.2) dan cotas inferiores incondicionales. Para paridad son ajustadas (2^{n-1}), pero la CNF requiere 2^{n-1} cláusulas. Para mayoría la cota es $2^{\Omega(n)}$, pero la CNF por minterms requiere $2^{\Theta(n)}$ cláusulas.
- (4) **Paridad como testigo del gap circuito/DSOP:** paridad tiene circuitos de tamaño $O(n)$ y $\chi_{\sqcup} = 2^{n-1}$, mostrando que el gap entre representación por circuitos y DSOP puede ser exponencial. La cuestión es si existe un gap análogo entre CNF (restringida a forma normal conjuntiva) y DSOP.
- (5) **Distinción clave:** el Corolario 2.6 acota la fragmentación **algorítmica** S_{peak} para CNFs **insatisfacibles** (PHP: la cota volumétrica da solo la constante 4, y la fragmentación exponencial queda como problema abierto; Tseitin: problema abierto). El Problema 9.6 concierne a CNFs **satisfacibles** y al mínimo **combinatorio** $\chi_{\sqcup}$, no al algorítmico. Para F insatisfacible, $\chi_{\sqcup}(\Omega_n) = 1$ trivialmente.

5.3 Problema abierto reformulado

En términos del mapa de compilación:

Problema abierto. ¿Existe una familia de DNFs $\{g_n\}$ de $m_n = \text{poly}(n)$ términos tal que toda DSOP de g_n requiera $2^{\Omega(n)}$ términos?

Incondicionalmente, el problema permanece abierto. La técnica espectral acota $\chi_{\sqcup}$ inferiormente por coeficientes de Fourier; para CNFs polinomiales generales los coeficientes de $\mathbf{1}_{U(F)}$ no se conocen en forma cerrada. La desigualdad con resolución arbórea (Teorema 2.3) y el contraejemplo (Proposición 2.4) muestran que las cotas de resolución arbórea no se transfieren a S_{part} ; el Corolario 2.6 proporciona cotas para S_{peak} vía S_{part} (PHP, cota volumétrica constante) pero no para $\chi_{\sqcup}$.

6 Conclusión

Se han establecido tres resultados que extienden el marco COVERTRACE-SAT:

- (1) **Desigualdad partición–resolución arbórea y contraejemplo** (Teorema 2.3, Proposición 2.4, Corolario 2.6): $S_{\text{part}}(F) \leq S_{\text{tr}}(F)$, con igualdad falsa en general (CNF explícita sobre 4 variables con $6 < 7$, árbol refutante de 7 hojas verificado). Para COVERTRACE sobre PHP_n^{n+1} la cota volumétrica directa da solo la constante 4 (cláusulas de hoyo de

ancho 2): la fragmentación exponencial en PHP y la transferencia de cotas de resolución arbórea para Tseitin quedan como problemas abiertos.

- (2) **Cota espectral de Fourier–Walsh** (Teorema 3.2): $\chi_{\sqcup}(S) \geq 2^d \cdot \max_{|\alpha|=d} |\widehat{1_S}(\alpha)|$. Ajustada para paridad, complementada por cota volumétrica para mayoría.
- (3) **Certificado poliédrico y codificación exacta** (Teoremas 4.1–4.3, Proposición 4.4): K desigualdades lineales describen $\text{Mod}(F)$ módulo integralidad (sin usar la disyuntividad, Observación 4.2); la codificación $\phi_n(s)_j = [s \in Q(u_j)]$ da la identidad exacta $\kappa_n(s) = [s \in U(F)]$; la decisión asociada es polinomial con entrada explícita y equivalente a SAT con entrada CNF (sin reducción de Karp).

El **Problema 9.6** permanece abierto. Equivale a separar CNF de DSOP en el mapa de compilación de conocimiento. Las técnicas espectral y volumétrica desarrolladas proporcionan herramientas de ataque pero no resuelven el problema.

□

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- *A Unique Encoding of SAT Assignments; Generalization of the SAT Equation via Bitwise OR*.
- *COVERTRACE-SAT as Disjoint-Subcube Knowledge Compilation* (Def. 9.1, Lema 3.2, Teo. 4.1, Teo. 6.2, Teo. 7.4, Prop. 9.4, Lema 9.2, Problema 9.6).
- *Teoría de Conservación de Óptimos y Complejidad* (Def. 4.1, §7, Teo. 9.5; citada solo como trasfondo: sus resultados no se invocan en esta Parte).

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